Gazi Shoaib

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Research Interests

Evaluation methodology and benchmark reliability in NLP; low-resource language processing, particularly Bangla; interpretable machine learning for health applications.

Education

M.Sc. in Applied Statistics and Data Science May 2024 – Jan 2026
Jahangirnagar University, Savar, Dhaka CGPA 3.19 / 4.00

- Thesis: *Explainable Machine Learning for Mental Health Risk Prediction*. Interpretable gradient-boosted model (87% accuracy) with SHAP and LIME attribution; threshold optimised to minimise false negatives.

B.Sc. in Computer Science and Engineering May 2019 – Nov 2023
North South University, Dhaka CGPA 3.00 / 4.00

Research

Preprints and Manuscripts

- **Gazi, S.** (2026). "Indistinguishable at the Top: A Statistical Audit of the BLP-2025 Bangla Hate Speech Shared Task." *Preprint*, Zenodo. doi:10.5281/zenodo.22472234
  Code: github.com/gazishoaib33/blp25-audit
  – Showed the top five systems in all three subtasks are statistically indistinguishable, and trained ten systems to measure the pairwise discordance the argument depends on across 45 system pairs.
  – Estimated the annotation ceiling from reported inter-annotator agreement and *rejected* the hypothesis that label noise explains the observed performance plateau.
  – Demonstrated that the task's micro-F1 metric cannot resolve its rarest class: at 29 of 10,200 test items, the metric's maximum response falls below the sampling noise floor.

Research Experience

Research and Development Intern May 2024 – Nov 2024
Penta Global Ltd., Dhaka

- Built NLP pipelines for sentiment analysis, named entity recognition, and topic modelling over real-world Bangla and English text.
- Trained and evaluated classification models; documented experimental protocol and presented findings to the technical team.

Selected Technical Projects

- **Natural-language-to-SQL generation with large language models.** FastAPI service using few-shot prompting and schema injection across three domain schemas; approximately 79% execution accuracy on the Spider benchmark at under 900 ms latency.
- **Automated meteorological data pipeline.** Hourly ETL over live weather data for 12 Bangladeshi cities with a derived heat-index and risk-scoring layer, persisted to SQLite.
- **Credit risk classification.** End-to-end pipeline (85% accuracy) with feature attribution identifying the dominant risk factors.

Technical Skills

- **Programming:** Python, SQL, C; Git, LaTeX, Jupyter, REST APIs, SQLite, MySQL, Power BI
- **Machine learning:** scikit-learn, XGBoost, TensorFlow; classification, regression, clustering, time-series forecasting; interpretability with SHAP and LIME
- **NLP and LLMs:** sentiment analysis, NER, topic modelling, prompt engineering, LLM API integration
- **Statistics:** hypothesis testing, confidence intervals, bootstrap resampling, McNemar's test, inter-annotator agreement (Fleiss' κ)

Languages and Additional Training

Languages: Bangla (native); English (professional working proficiency).
Training: Machine Learning Specialization (Coursera); Data Analytics Career Track (Bohubrihi); Data Science Course (NSU

ACM).